EXP NO:6 Time Series Analysis of Website Traffic

AIM:
To perform time series analysis of website traffic.

PROGRAM:

python

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Creating daily website traffic data for 180 days
np.random.seed(42)
dates = pd.date_range(start='2024-01-01', periods=180, freq='D')
days = np.arange(180)

visitors = (
    1000
    + days * 5
    + 200 * np.sin(days * 2 * np.pi / 7)
    + np.random.normal(0, 50, 180)
)

df = pd.DataFrame({'Date': dates, 'Visitors': visitors.astype(int)})
df.set_index('Date', inplace=True)
df.head()

# Information about the dataset
df.info()

# Statistical description
df.describe()

# Line plot of daily traffic
plt.figure(figsize=(10, 4))
plt.plot(df.index, df['Visitors'])
plt.title("Daily Website Traffic")
plt.xlabel("Date")
plt.ylabel("Visitors")
plt.grid(True)
plt.show()

# 7-day rolling average (trend)
df['Rolling_7'] = df['Visitors'].rolling(window=7).mean()

plt.figure(figsize=(10, 4))
plt.plot(df.index, df['Visitors'], label='Daily Visitors', alpha=0.5)
plt.plot(df.index, df['Rolling_7'], label='7-day Average', color='red')
plt.title("Website Traffic with 7-day Rolling Average")
plt.xlabel("Date")
plt.ylabel("Visitors")
plt.legend()
plt.show()

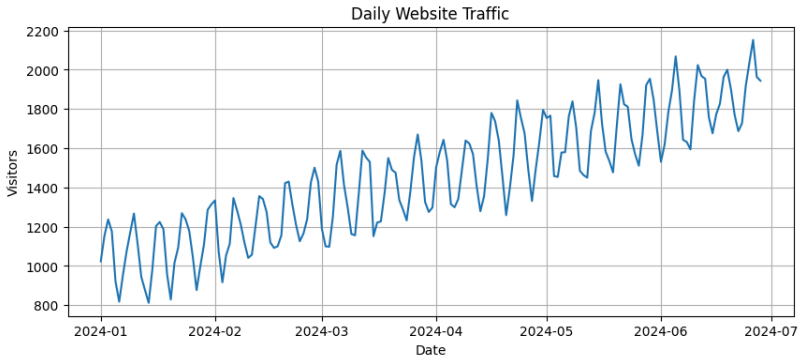
# Weekly average traffic
weekly = df['Visitors'].resample('W').mean()
weekly.head()

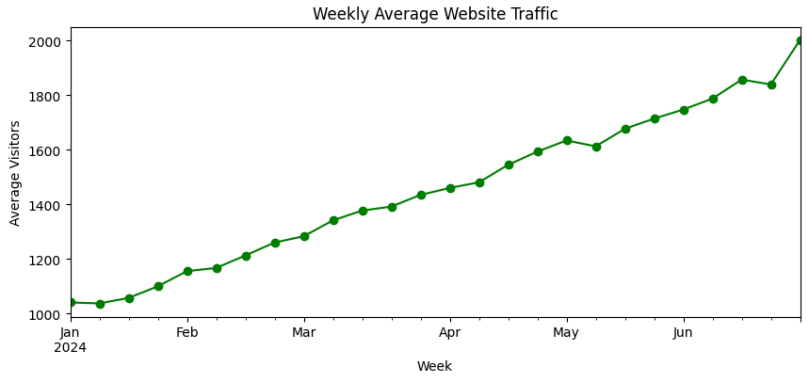
plt.figure(figsize=(10, 4))
weekly.plot(marker='o', color='green')
plt.title("Weekly Average Website Traffic")
plt.xlabel("Week")
plt.ylabel("Average Visitors")
plt.show()

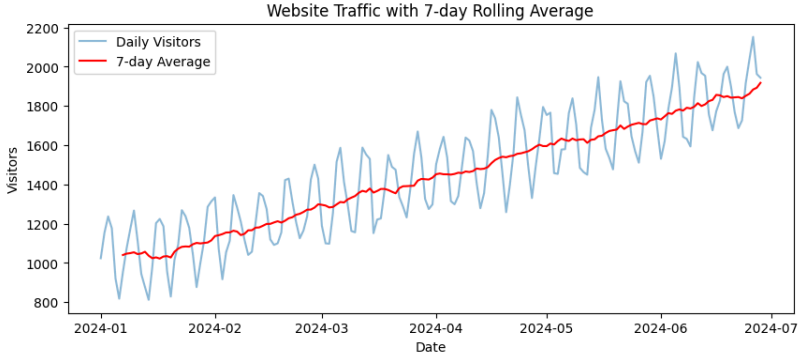
# Average traffic for each day of the week (seasonality)
df['Day'] = df.index.day_name()
day_avg = df.groupby('Day')['Visitors'].mean().reindex(
    ['Monday', 'Tuesday', 'Wednesday', 'Thursday',
     'Friday', 'Saturday', 'Sunday']
)
print(day_avg)

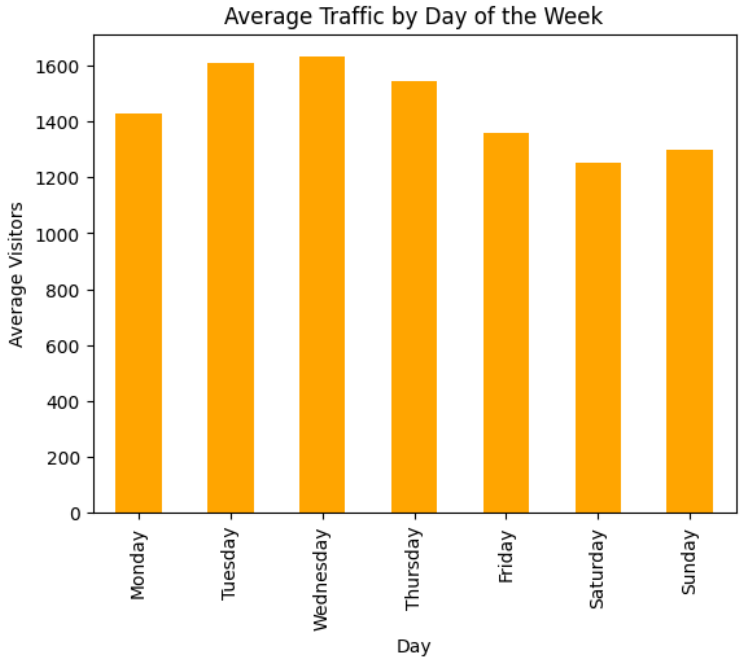
day_avg.plot(kind='bar', color='orange')
plt.title("Average Traffic by Day of the Week")
plt.ylabel("Average Visitors")
plt.show()
```

OUTPUT:


Daily Website Traffic
Visitors
Date
2200
2000
1800
1600
1400
1200
1000
800
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07


Weekly Average Website Traffic
Average Visitors
Week
2000
1800
1600
1400
1200
1000
Jan
2024
Feb
Mar
Apr
May
Jun


Website Traffic with 7-day Rolling Average
Daily Visitors
7-day Average
Visitors
Date
2200
2000
1800
1600
1400
1200
1000
800
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07




```
Day
Monday       1427.538462
Tuesday      1607.384615
Wednesday    1630.769231
Thursday     1544.846154
Friday       1358.461538
Saturday     1253.560000
Sunday       1299.080000
Name: Visitors, dtype: float64
```

RESULT:

Thus the program to perform time series analysis of website traffic was executed successfully.